MILITARY SPECIFICATION

CONTAINER, AMMUNITION, PLASTIC

1. SCOPE

1.1 Scope. This specification covers molded, high density polyethylene plastic containers, for shipment and storage of ammunition.

1.2 Classification.-The plastic ammunition containers shall be of the following types as specified:

Type I - For 81mm Mortar Ammunition
Type II - For 90mm Recoilless Rifle Ammunition
Type III - For 105mm Howitzer Ammunition
Type IV - For 105mm Gun Ammunition
Type V - For 107mm Mortar Ammunition

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-640 - Boxes, Folding, Fiberboard, Corrugated, Triple Wall

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-P-10025 - Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

FSC: 8140

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-129 - Marking for Shipment or Storage.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

ORDJU-4021 - Unit Load (strapped) for Shipping and Storage of 81mm Mortar Ammunition Packed 4 in Wirebound Box
19-48-4020-1-2-5-11PA 1000 - Carloading, Truckloading and Storage of Detailed Palletized (Strapped) Unit Loads of Boxed Ammunition and Components.
8882049 - Crate, Wrap Around, for Container, Ammo Plastic, M557 for Cartridge, 90mm, HEAT and Practice, M371 and M371E1 for Rifle, M67
8837375 - Picatinny Arsenal Environmental Testing Criteria for Shipping Containers
8861446 - Container, Plastic, Ammo, [illegible] for Cartridge, 105mm, APDS-T, M392A1 for Gun M68
8861659 - Container, Ammo, Plastic [illegible] for Cartridge, 105mm, HE, M482
8861923 - Container, Ammo, Plastic, [illegible] for Cartridge, 105mm, HEAT-T and TP-T for Gun, M68
8862149 - Container, Ammo, Plastic, [illegible] for Cartridge, 105mm, HEP-T, M393A1 for Gun, M68

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8862317 - Container, Ammo, Plastic, M557 for Cartridge 90MM, HEAT, M371 for Rifle M67
8864657 - Container, Ammo. Plastic, M513 for 81MM, Mortar Ammunition
8864663 - Crate, Wrap Around, for Container, Ammunition, High Density Polyethylene, for 81MM Mortar Ammunition
8875970 - Container, Ammo, Plastic for Cartridge, 107MM, XM453
8885050 - Container, Ammo, Plastic, for Cartridge, Illuminating, XM512E1
9209385 - Container, Ammo, Plastic, for Cartridge, 105MM, RAP, XM548
9209387 - Crate, Wood, Open, Wirebound for 105MM (How.) Ammo, in Plastic Ctr
9207578 - Box, Wirebound, Packing, Ammunition, for 105MM, Ammunition in Plastic Container,

PUBLICATIONS

ARMY

ET-8861446 - Equipment Tabulation
ET-8861659 - Equipment Tabulation
ET-8861923 - Equipment Tabulation
ET-8862149 - Equipment Tabulation
ET-8862317 - Equipment Tabulation
ET-8864657 - Equipment Tabulation
ET-8875970 - Equipment Tabulation
ET-8885050 - Equipment Tabulation
ET-9209385 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials Publications;

D1238-57T - Measuring Flow Rate of Thermoplastics by Extrusion Plastometer
D1505-57T - Measurement of Density of Plastics by the Density Gradient Technique

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

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3. REQUIREMENTS

3.1 Material.-The resin used in the manufacture of the container shall be high density polyethylene or copolymer thereof, which contains 0.3 percent, plus or minus 0.05 percent carbon black, an anti-oxident and an ultraviolet stabilizer uniformly dispersed. In addition, the containers shall be in accordance with the applicable drawings and specifications.

3.2 Mold release agent.-No release agent used in the molding process shall have silicone components, or other components, that may adversely affect the printability of the exterior surfaces of the container.

3.3 Physical properties.-The high density polyethylene container shall conform to the physical properties listed in Table I when tested as specified in 4.3.

3.4 Reworked material.-No reworked material or reground scrap (including runners and sprues) shall be utilized in molding containers under this specification. However, initial pelletizing of the molding compositions used shall not be construed as reworking.

3.5 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of cracks, burrs, surface defects, flash air bubbles, holes, blisters, amides, dirt, twists, and scrap. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. (see 4.4.7).

TABLE I
Physical Properties

Test	Requirements
Melt Index, grams/10 minutes with 2160g load	3.0 max.
Density, gms/ml	0.955 min.
Color Number (Fed Std 595)	All sections of the container shall closely approximate the composition color number indicated on the applicable drawing. Homogeneity of color within a lot shall be maintained.

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Thermal stress cracking resistance	The containers shall not crack, split, warp, or break when exposed for 100 hours at 71 degrees C. and tested as specified in 4.4.3.
Torque test	The closure shall be completely removed without exceeding the maximum torque specified on the applicable drawing when tested as specified in 4.4.4
Free fall drop test 23 degrees Centigrade (°C.) 71 degrees C. -40 degrees C.	The container shall withstand the free fall drop test without evidence of cracking, splitting, breaking or damage to item packed therein. Closures shall remain intact and closure gaskets unbroken when tested as specified in 4.4.5.
Wettability of container	The prepared exterior surface of the container shall hold water as a continuous film for a minimum of two seconds, when tested as specified in 4.4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services

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conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

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4.2 Initial production inspection (see 6.5)

4.2.1 Submission.-Prior to the initiation of regular production the contractor shall submit an initial production sample consisting of fifty components from each die or mold, which shall be selected proportionally from each cavity, which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the Contracting Officer. The component shall be identified by die or mold number, and the cavity number in the die or mold, as applicable. All parts and materials shall be obtained from the same source of supply as will be used in regular production. The initial production sample shall be accompanied by certificates of conformance for all materials. In addition, the contractor shall furnish a statement of his process variables such as temperatures, pressures and cycles. A sample shall also be submitted whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Assemblies and components shall be subjected by the Government to any or all of the inspections and tests specified in this specification. In addition, one component from each cavity of each die or mold, as applicable, shall be inspected for compliance with any or all of the requirements specified on the applicable drawing.

4.2.3 Rejection.-If any assembly or component fails to comply with any or all of the applicable requirements, the initial production sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit new or supplemental quantities until such time as an acceptable sample is submitted, or until otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of high density polyethylene plastic molded containers produced by one manufacturer in one unchanged continuous process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, interchangeability or storage, as determined by the Government shall not necessitate changing the lot interfix number.

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The examinations and tests listed herein shall be performed on inspection lots submitted in accordance with Standard MIL-STD-105 or MIL-STD-1235, as applicable. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168. Each lot shall contain:

a. Material from one lot from the same manufacturer.
b. Material of one color designation.
c. Components from the same manufacturer.

4.3.2 Examination.-Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with Standard MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 - Type I (see 4.3.3.2)

4.3.2.1.1 Closure (see dwgs. 8864659 and 8882223 covering a detail of dwgs. 8864657 and 8885050).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Depth of thread, min.	Gage	01001
102.	Width of slot (s)	Gage	01002
103.	Inside diameter	Gage	01003
104.	Location of threads from face of open end	Gage	01004
105.	Wall thickness, min.	Gage	01005
106.	Pitch location	Gage	01006
107.	Width of thread at pitch location	Gage	01007
108.	Chamfer at open end missing	Visual	01008
Minor:	AQL 2.50 percent		
201.	Large outside diameter	Gage	01009
202.	Total length	Gage	01010
203.	Thickness of base	Gage	01011

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204.	Internal radius missing.....	Visual	01012
205.	Chamfer at closed end missing.	Visual	01013
206.	Color incorrect.....	Visual	01014
207.	Evidence of poor workmanship (see 3.5).....	Visual	01015

4.3.2.1.2 Tube (see dwg. 8864658 covering a detail of dwgs. 8864657 and 8885050)

Categories	Defects	Method of Inspection	Code No.
Major:	AQL 1.50 percent		
101.	Pitch location.....	Gage	02001
102.	Depth of thread, min.....	Gage	02002
103.	Wall thickness, min.....	Gage	02003
104.	Length of thread, min.....	Gage	02004
105.	Location of thread from open end face.....	Gage	02005
106.	Depth of "O" ring groove.....	Gage	02006
107.	Straightness.....	Gage	02007
108.	Inside diameter at open end....	Gage	02008
109.	Location of "O" ring seat from open end face.....	Gage	02009
110.	Diameter at fuze cavity opening.	Gage	02010
111.	Depth of fuze cavity.....	Gage	02011
112.	Length of thread engagement, min.....	Gage	02012
113.	Width of thread at pitch location.....	Gage	02013
Minor:	AQL 1.50 percent		
201.	Outside diameter at "O" ring seat.....	Gage	02014
202.	Total length.....	Gage	02015
203.	Incorrect color.....	Visual	02016
204.	Chamfer at closed end missing..	Visual	02017
205.	Evidence of poor workmanship (see 3.5).....	Visual	02018

4.3.2.1.3 Crate, wrap around (see dwg. 8864663).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Board broken or split.....	Visual	03001
102.	Wire missing, broken or loose..	Visual	03002
103.	Incorrect packing material (see 5.1.1).....	Visual	03003

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Minor:	AQL 1.00 percent		
201.	Loops improperly engaged.....	Visual	03004
202.	Staples missing, broken or loose..	Visual	03005

4.3.2.1.4 Unit load (strapped) (see dwg. ORDJU-4021)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Pallet damaged.....	Visual	04001
102.	Strapping missing, loose or obviously mislocated.....	Visual	04002
Minor:	AQL 1.00 percent		
201.	Number of containers incorrect....	Visual	04003
202.	Strapping improperly engaged.....	Visual	04004

4.3.2.2 - Type II (see 4.3.3.2)

4.3.2.2.1 Closure (see dwg. 8861666 covering a detail of dwg. 8862317)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Location of shoulder from face of open end.....	Gage	01001
102.	Length of thread from face of open end.....	Gage	01002
103.	Location of thread from face of open end.....	Gage	01003
104.	Depth of thread, min.....	Gage	01004
105.	Perpendicularity of face at open end with minor diameter of threads.....	Gage	01005
106.	Wall thickness, min.....	Gage	01006
107.	Depth of cavity from face at open end.....	Gage	01007
108.	Pitch location.....	Gage	01008
109.	Width of thread at pitch location.	Gage	01009
110.	Chamfer at open end missing.....	Visual	01010

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 2.50 percent		
201.	Large outside diameter	Gage	01011
202.	Total length	Gage	01012
203.	Chamfer at closed end missing	Visual	01013
204.	Color incorrect	Visual	01014
205.	Evidence of poor workmanship (see 3.5)	Visual	01015

4.3.2.2.2 Body (see dwg. 8861665 covering a detail of dwg. 8862317)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.50 percent		
101.	Location of thread from face of open end	Gage	02001
102.	Length of thread from face of open end	Gage	02002
103.	Location of "O" ring seat from face of open end	Gage	02003
104.	Straightness	Gage	02004
105.	Perpendicularity of "O" ring seat with major diameter of threads	Gage	02005
106.	"O" ring groove radii	Gage	02006
107.	Wall thickness, min	Gage	02007
108.	Large inside diameter of body cavity	Gage	02008
109.	Small inside diameter of nose cavity	Gage	02009
110.	Intermediate inside diameter of body cavity	Gage	02010
111.	Small inside diameter of body cavity	Gage	02011
112.	Location of large internal chamfer	Gage	02012
113.	Depth of internal cavities from face of open end	Gage	02013
114.	Length of thread engagement, min.	Gage	02014
115.	Pitch location	Gage	02015
116.	Width of thread at pitch location	Gage	02016
117.	Chamfer at face of open end missing	Visual	02017

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Minor:	AQL 1.50 percent		
201.	Outside diameter at "O" ring seat	Gage	02016
202.	Total length	Gage	02017
203.	Incorrect color	Visual	02018
204.	Evidence of poor workmanship (see 3.5)	Visual	02019

4.3.2.2.3 Crate, wrap around (see dwg. 8882049)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Board broken or split	Visual	03001
102.	Hardware or wire missing, broken or loose	Visual	03002
103.	Incorrect packing material (see 5.1.1)	Visual	03003
Minor:	AQL 0.65 percent		
201.	Hardware or wire loops improperly engaged	Visual	03004

4.3.2.2.4 Unit load (strapped) (see Unit 9, Page 28, Dwg. 19-48-4020-1-2-5-11PA 1000)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Pallet damaged	Visual	04001
102.	Strapping missing, loose or obviously mislocated	Visual	04002
Minor:	AQL 1.00 percent		
201.	Number of containers incorrect	Visual	04003
202.	Strapping improperly engaged	Visual	04004

4.3.2.3 Type III (see 4.3.3.2)

4.3.2.3.1 Closure (see dwgs. 8861663 and 9209386 covering a detail of dwgs. 8861659 and 9209385).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Location of thread from open end face	Gage	01001
102.	Length of thread, min	Gage	01002
103.	Depth of thread, min	Gage	01003
104.	Width of slot (s)	Gage	01004
105.	Inside diameter of closed end	Gage	01005
106.	Pitch location	Gage	01006
107.	Width of thread at pitch location	Gage	01007
Minor:	AQL 2.50 percent		
201.	Total length	Gage	01008
202.	Outside diameter at open end	Gage	01009
203.	Chamfer at closed end missing	Visual	01010
204.	Color incorrect	Visual	01011
205.	Evidence of poor workmanship (see 3.5)	Visual	01012

4.3.2.3.2 - Aft Section (see dwg. 8861661 covering a detail of dwgs. 8861659 and 9209385)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Straightness	Gage	02001
102.	Length of rearward thread, min.	Gage	02002
103.	Location of "O" ring seat from rearward face	Gage	02003
104.	Depth of thread, min	Gage	02004
105.	Location of packing stop from forward face	Gage	02005
106.	Length of forward thread, min.	Gage	02006
107.	Inside diameter, min	Gage	02007
108.	Length of thread engagement, min	Gage	02008

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109.	Pitch location	Gage	02009
110.	Width of thread at pitch location	Gage	02010
Minor:	AQL 1.50 percent		
201.	Large outside diameter	Gage	02011
202.	Total length	Gage	02012
203.	Color incorrect	Visual	02013
204.	Evidence of poor workmanship (see 3.5)	Visual	02014

4.3.2.3.3 - Forward Section (see dwg. 8861662 covering a detail of dwgs. 8861659 and 9209385)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Straightness	Gage	03001
102.	Length of external thread, min.	Gage	03002
103.	Location of "O" ring seat from face at projectile end	Gage	03003
104.	Depth of thread, min	Gage	03004
105.	Length of internal thread, min.	Gage	03005
106.	Location of internal thread from face at cartridge case end	Gage	03006
107.	Small internal diameter	Gage	03007
108.	Pitch location	Gage	03008
109.	Width of thread at pitch location	Gage	03009
Minor:	AQL 1.50 percent		
201.	Large outside diameter	Gage	03010
202.	Total length	Gage	03011
203.	Color incorrect	Visual	03012
204.	Evidence of poor workmanship (see 3.5)	Visual	03013

4.3.2.3.4 - Tube Protector (see dwg. 8861664 covering a detail of dwg. 8861659).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined.		
Minor:	AQL 2.50 percent		
201.	Length	Gage	04001
202.	Wall thickness	Gage	04002
203.	Outside diameter	Gage	04003
204.	Diameter of hole	Gage	04004
205.	Evidence of poor workmanship (see 3.5)	Visual	04005

4.3.2.3.5 Unit Load (see dwg. 9209387).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Pallet damaged	Visual	05001
102.	Strapping missing, broken or obviously mislocated	Visual	05002
Minor:	AQL 1.00 percent		
201.	Number of containers incorrect	Visual	05003
202.	Strapping improperly engaged	Visual	05004

4.3.2.4 - Type IV (see 4.3.3.2).

4.3.2.4.1.-Closure (see dwg. 8861448 covering a detail of dwgs. 8861446, 8861923 and 8862149).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Depth of thread, min	Gage	01001
102.	Width of slot (s)	Gage	01002

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103.	Location of thread from face of open end	Gage	01003
104.	Length of thread from face of open end	Gage	01004
105.	Perpendicularity of face of open end with inside diameter	Gage	01005
106.	Inside diameter	Gage	01006
107.	Wall thickness, min	Gage	01007
108.	Pitch location	Gage	01008
109.	Width of thread at pitch location	Gage	01009
110.	Chamfer at face of open end missing	Visual	01010
Minor:	AQL 2.50 percent		
201.	Large outside diameter	Gage	01011
202.	Total length	Gage	01012
203.	Thickness of base	Gage	01013
204.	Chamfer at closed end missing	Visual	01014
205.	Color incorrect	Visual	01015
206.	Evidence of poor workmanship (see 3.5)	Visual	01016

4.3.2.4.2 - Center Tube (see dwg. 8861450 covering a detail of dwgs. 8861446, 8861923 and 8862149).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.50 percent		
101.	Concentricity of major diameter of external thread with diameter of external ribs	Gage	02001
102.	Perpendicularity of shoulder at forward end with major diameter of threads	Gage	02002
103.	Location of thread, min. at forward end	Gage	02003
104.	Length of thread from face at forward end	Gage	02004
105.	Location of shoulder at forward end	Gage	02005

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106.	Perpendicularity of "O" ring seat with diameter over external ribs ...	Gage	02006
107.	Location of "O" ring seat from face at rearward end	Gage	02007
108.	Length of thread, min., from face at rearward end	Gage	02008
109.	Location of thread from face at rearward end	Gage	02009
110.	Inside diameter over ribs	Gage	02010
111.	Concentricity of ribs with outside diameter	Gage	02011
112.	Wall thickness	Gage	02012
113.	Length of thread, min.	Gage	02013
114.	Length of thread engagement, min. ..	Gage	02014
115.	Pitch location	Gage	02015
116.	Width of thread at pitch location	Gage	02016
117.	Chamfer at rearward end missing	Visual	02017
Minor:	AQL 2.50 percent		
201.	Large outside diameter	Gage	02018
202.	Total length	Gage	02019
203.	Internal rib missing	Visual	02020
204.	Color incorrect	Visual	02021
205.	Evidence of poor workmanship (see 3.5)	Visual	02022

4.3.2.4.3 - Nose Section (see dwgs. 8861449, 8861922 and 8862150 covering a detail of dwgs. 8861446, 8861923 and 8862149).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Location of thread from face at open end	Gage	03001
102.	Length of thread from face at open end	Gage	03002
103.	Minor diameter of threads	Gage	03003

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104.	Perpendicularity of face at open end with minor diameter of threads	Gage	03004
105.	Wall thickness, min	Gage	03005
106.	Small inside diameter	Gage	03006
107.	Intermediate inside diameter	Gage	03007
108.	Pitch location	Gage	03008
109.	Width of thread at pitch location	Gage	03009
110.	Chamfer at face of open end missing	Visual	03010
Minor:	AQL 2.50 percent		
201.	Large outside diameter	Gage	03011
202.	Total length	Gage	03012
203.	Location of internal ribs from open end face (as applicable)	Gage	03013
204.	Color incorrect	Visual	03014
205.	Evidence of poor workmanship (see 3.5)	Visual	03015

4.3.2.4.4 - Insert (see dwg. 9211370 covering a detail of dwg. 8861923).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Large outside diameter, min	Gage	04001
102.	Small outside diameter, min	Gage	04002
103.	Intermediate outside diameter, min	Gage	04003
104.	Location of flange	Gage	04004
Minor:	AQL 2.50 percent		
201.	Thickness of flange	Gage	04005
202.	Inside diameter	Gage	04006
203.	Total length	Gage	04007
204.	Rib missing or broken	Visual	04008
205.	Evidence of poor workmanship (see 3.5)	Visual	04009

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4.3.2.4.5 - Unit load (see dwg. 9207578)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Pallet damaged	Visual	05001
102.	Strapping missing, loose or obviously mislocated	Visual	05002
Minor:	AQL 1.00 percent		
201.	Number of containers incorrect	Visual	05003
202.	Strapping improperly engaged	Visual	05004

4.3.2.5 - Type V (see 4.3.3.2).

4.3.2.5.1 - Closure (see dwg. 8875972 covering a detail of dwg. 8875970).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Location of thread from open end face	Gage	01001
102.	Length of thread, min	Gage	01002
103.	Straightness	Gage	01003
104.	Pitch location	Gage	01004
105.	Depth of thread	Gage	01005
106.	Major diameter of threads	Gage	01006
107.	Intermediate inside diameter	Gage	01007
108.	Small inside diameter	Gage	01008
109.	Wall thickness	Gage	01009
110.	Width of slot (s)	Gage	01010
111.	Width of threads at pitch location	Gage	01011
112.	Chamfer at open end missing	Visual	01012

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Minor:	AQL 1.50 percent		
201.	Total length	Gage	01013
202.	Large outside diameter	Gage	01014
203.	Color incorrect	Visual	01015
204.	Evidence of poor workmanship (see 3.5)	Visual	01016

4.3.2.5.2 - Body (see dwg. 8875971 covering a detail of dwg. 8875970).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.50 percent		
101.	Pitch location	Gage	02001
102.	Depth of thread, min	Gage	02002
103.	Wall thickness	Gage	02003
104.	Location of thread from open end face	Gage	02004
105.	Length of thread, min	Gage	02005
106.	Location of "O" ring seat from open end face	Gage	02006
107.	Inside diameter at open end	Gage	02007
108.	Large internal diameter of nose cavity	Gage	02008
109.	Small diameter of nose cavity	Gage	02009
110.	Depth of nose cavity from shoulder	Gage	02010
111.	Depth of seal groove	Gage	02011
112.	Straightness	Gage	02012
113.	Length of thread engagement, min	Gage	02013
114.	Width of thread at pitch location	Gage	02014
Minor:	AQL 1.50 percent		
201.	Large outside diameter	Gage	02015
202.	Total length	Gage	02016
203.	Incorrect color	Visual	02017
204.	Evidence of poor workmanship (see 3.5)	Visual	02018

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4.3.2.5.3 - Unit load (see dwg. 9207578)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Pallet damaged	Visual	03001
102.	Strapping missing, loose or obviously mislocated	Visual	03002
Minor:	AQL 1.00 percent		
201.	Number of containers incorrect.	Visual	03003
202.	Strapping improperly engaged	Visual	03004

4.3.3 Tests

4.3.3.1 Classification of tests.-The examination and testing of the container assemblies shall be classified as follows: (see 6.7)

(a) Initial production tests

(b) Regular production tests

4.3.3.1.1 Initial production tests.-The initial production tests shall consist of all inspections tests and quantities as specified in 4.3 except that a quantity of ten container assemblies will be subjected to the tests specified in 4.3.3.8.

4.3.3.1.2 Regular production tests.-The regular production test for acceptance of individual lots shall consist of inspections and tests as specified in 4.3.

4.3.3.2 Delay time in dimensioning examinations.-No attempt shall be made to determine compliance with dimension and tolerance requirements of the sections for a minimum period of 48 hours after molding.

4.3.3.3 Melt index (see Table 1) Major Defect, Code No. 10001. Three determinations shall be made and averaged from each lot for this test. Failure of average to meet the values when tested as specified in 4.4.1 shall be cause for the lot to be rejected.

4.3.3.4 Density (see Table 1) Major Defect, Code No. 10002.

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Three determinations shall be made and averaged from each lot for this test. Failure of average to meet the values when tested as specified in 4.4.2 shall be cause for the lot to be rejected.

4.3.3.5 Thermal stress cracking resistance (see Table 1) Major defect, Code No. 10003.-A sample of five container assemblies shall be selected at random from each lot for this test. If one or more assembly fails to comply with the requirements when tested as specified in 4.4.3, the lot shall be rejected.

4.3.3.6 Torque test (see Table 1) - Major defect, Code No. 10004.-Determination of compliance with this requirement shall be made on the samples required by 4.3.3.5. Torque values of five containers shall be averaged. If the average exceeds the maximum torque requirement when tested as specified in 4.4.4, the lot shall be rejected.

4.3.3.7 Free fall drop test (see Table 1) - Major Defect, Code No. 10005.-A sample of five container assemblies shall be selected at random from each lot for this test. If one or more assembly fails to comply with the requirement when tested as specified in 4.4.5, the lot shall be rejected.

4.3.3.8 Wettability of container (see Table 1) - Major defect, Code No. 10006.-A sample of twenty-five container assemblies shall be selected at random from each eight hours production and tested as specified in 4.4.6. If one or more container fails to comply with the requirement specified, the entire quantity produced on initial production or since the last acceptance test shall be tested and the defective containers shall be rejected and removed from the lot. The reject containers shall be reprocessed and retested 100 percent. Any container failing to meet the specified requirement on retest shall be removed from the lot and shall not be eligible for further acceptance.

4.3.3.9 Void determination.- Major defect, Code No. 10007.- A sample of five of each component, representative of each shipment shall be selected at random for this inspection. If one or more components fails to comply with the requirement when tested as specified in 4.4.7, the lot shall be rejected.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-8861446, 8861659, 8861923, 8862149, 8862317, 8864657, 8875970, 8885050 and 9209385 identifies the inspection equipment required

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to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for deisgn, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 Melt Index.-Melt Index shall be determined in accordance with ASTM D1238-57T, Condition E. The test specimens shall be cut from a container.

4.4.2 Density.-The density of the specimens shall be determined in accordance with ASTM D1505-57T. The test specimen shall be cut from a container. Specimens shall not be annealed or otherwise treated so as to change the crystallinity. (see 6.6)

4.4.3 Thermal stress cracking resistance.-The closure shall be assembled to the tube or body, without closure sealing gasket, to within approximately 1/8 inch of the O ring seat. The assemblies shall then be subjected to 100 hours at 71 degrees Centigrade, plus or minus 1 degree C. Immediately following the 100 hour time period each container shall be removed individually and without disturbing the closure position, shall be visually examined while

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warm (see 3.4). The container assemblies shall then be subjected to the procedure of 4.4.4. Following that procedure the containers shall each be cut open longitudinally and the interior surfaces examined visually for evidence of cracks, splits, warping or other deformation.

4.4.4 Torque Test.-Following the initial visual examination of 4.4.3 the containers shall be allowed to come to temperature equilibrium at 23 degrees Centigrade, plus or minus 1 degree C. While at that temperature the amount of torque required to completely remove each closure shall be determined using inspection equipment in accordance with 4.3.4.

4.4.5 Free Fall Drop Test

4.4.5.1 Individual containers without over pack shall be prepared in accordance with the loading, assembly and packing specification for the ammunition item being packed, except that the rounds shall be completely inert.

Using inspection equipment in accordance with 4.3.4, the assemblies shall be subjected to the free fall drop test sequence for cylindrical containers indicated in Picatinny Arsenal Environmental Testing Criteria for Shipping Containers, Drawing 8837375, except that for the edge drops a free fall angle of 10.5 plus or minus 1 degree shall be maintained between the axis of the container and the vertical.

Following the falls the containers shall be opened and examined for evidence of cracking, splitting or breaking of tube or closure. The closures shall have remained intact prior to opening for examination and the gaskets remained unbroken. The containers shall be cut open longitudinally and the interior surfaces similarly examined for such damage.

Additionally, the ammunition item being packed shall be examined for dented or ruptured cartridge case, loosening of the crimp of the Cartridge to the projectile (as applicable), borken propelling charge bag, damaged rotating band or fuze, or any other damage which would impair the functionality of the ammunition. In case of dispute, the ammunition shall be chamber gaged or fitted into the appropriate weapon to determine its functionality. Failure in this latter examination shall be cause for rejection of the container.

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4.4.5.2 Additional containers shall be prepared as in 4.4.5.1 and then conditioned at 71 degrees Centigrade plus or minus 1 degree C. for 24 hours. The 13 falls shall be completed within 5 minutes after removal from the conditioning chamber in order to maintain as much as possible of the high temperature in the test assemblies. Following the free falls the containers shall be examined as in 4.4.5.1.

4.4.5.3 Additional containers shall be prepared as in 4.4.5.1 and then conditioned at -54.3 degrees Centigrade plus or minus 1 degree C. for 8 hours. The containers shall then be conditioned for 16 hours (overnight) at -40 degrees C. plus or minus 1 degree C. and then drop tested as specified in 4.4.5.1. The 13 falls shall be completed within 5 minutes after removal from the conditioning chamber in order to maintain as much as possible of the low temperature in the test assemblies. Following the free falls the containers shall be examined as in 4.4.5.1.

4.4.6 Wettability of container.-The exterior surface of the container, treated by any standard commercial method (such as flame treatment or chromic acid treatment) for promoting printability of polyethylene surfaces, (and free from subsequent contamination through handling) shall be fully submerged in a water bath. With container held in a horizontal position as it is removed from the water, the tendency of the surface water film to coalesce shall be observed and timed in seconds.

4.4.7 Void determination.-The components selected for this inspection shall be cut into four 90 degree segments. Observation shall be made for indications of voids beyond the drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A.-The container assembly shall be packed in accordance with 5.1.1.1, 5.1.1.2 or 5.1.1.3, as specified by the procuring activity. Precaution shall be taken to prevent treated exterior surfaces from becoming contaminated by dirt, dust, light oil film deposits from the air or by handling during shipment and storage. Certain types of paper such as chipboard have an affinity for treated polyethylene surfaces and may destroy the effectiveness of the surface treatment. Therefore suitable means shall be provided to prevent direct contact of the containers with such packing materials.

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5.1.1.1 Types I & II.-Container assemblies, suitably protected from contamination as specified in 5.1.1 shall be packed in accordance with Dwg. 8864663 (Type I) or Dwg. 8882049 (Type II) with the exception that containers and crates shall not be marked and car seal not applied. Packed crates shall be palletized in accordance with Dwg. ORDJU-4021. (Type I) or Unit 9, Page 28, Dwg. 19-48-4020-1-2-5-11 PA 1000 (Type II).

5.1.1.2 Types III, IV & V.-Container assemblies, suitably protected from contamination as specified in 5.1.1 shall be unit loaded (palletized) in accordance with Dwgs. 9209387 (Type III), 9207578 (Type IV) and 9207578 (Type V), with the exception that containers shall not be marked and seals not applied.

5.1.1.3 Container assemblies, suitably protected from contamination as specified in 5.1.1 shall be packed in wire bound boxes in accordance with Class 3, Specification PPP-B-585. Weight limitations and closure requirement of Specification PPP-B-585 shall be observed.

5.2 Packing - Level C.

5.2.1 Packing shall be in accordance with Specification MIL-P-10025 with suitable protection from contamination provided as specified in 5.1.1.

5.3 Marking

5.3.1 Level A.-Level A shipments shall be marked in accordance with Standard MIL-STD-129 or as directed by the procuring activity.

5.3.2 Level C.-Level C shipments shall be marked in accordance with Specification MIL-P-10025.

6. NOTES

6.1 Ordering data.-Procurement documents shall specidy the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1)
c. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment
SCD - Specification control drawing.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their desigsn in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor desigsn.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608. and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo

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or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the IEL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the CJommanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Molding instructions.-The particular material selected as to density and melt index should be injection molded so as to meet all the requirements of this specification. It must be noted that selection of a material complying with the density and melt index requirements does not necessarily insure compliance with the performance requirements of the specification nor relieve the molder of his responsibility for molding in such a fashion that the containers will meet the requirement.

6.5 Submission of initial production samples.-Instructions, as to the location for the evaluation of the initial production sample shall be obtained from the Contracting Officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI.

6.6 Density specimens.-Saw speed in excess of 500 to 600 lineal feet per minute have been found to be unsatisfactory as excess speed can essentially melt the material and greatly change the density.

6.7 Classification of tests.-The term classification of tests as used in the text of this specification are intended to be used as follows:

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6.7.1 Initial production tests will be the examination and tests performed prior to the start of regular production to assure the Government that the methods and processes being used for regular production will produce an end product by the container supplier which will be acceptable under this specificatior.

6.7.2 Regular production tests will be the examination and tests performed on a lot to lot basis during regular production to assure the Government that the methods and processes being used for production will produce an end product which will be acceptable under this specification.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 8140-A019

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